Powering Tomorrow:
AI-Based Forecasting for Smart Energy Management

Technical Report
LUISS AI Techniques x Enel Global ICT Project Work

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1. Introduction

In recent years, energy consumption forecasting has become increasingly important for modern energy management systems. Accurate electric load prediction allows utility companies and energy providers to optimize resource allocation, improve operational efficiency and support more sustainable energy usage.

The objective of this project is to develop a time series forecasting pipeline capable of predicting household electricity consumption using historical power usage data. This project was carried out in collaboration with Enel Global ICT.

The analysis was conducted using the “Individual Household Electric Power Consumption” dataset from the UCI Machine Learning Repository. This dataset contains more than two million measurements collected between 2006 and 2010 from a household located near Paris, France. The data includes several variables related to electric consumption, voltage, current intensity and energy sub-metering.

The project focuses on analysing temporal patterns in electricity consumption, including trends, seasonality and fluctuations over time. In order to achieve this goal, we implemented a complete workflow including data loading, preprocessing, exploratory data analysis, forecasting model development and model evaluation.

Different forecasting approaches were explored and compared under the same evaluation framework. The implementation was organised into modular Python components to ensure readability, reproducibility and scalability of the experiments.

The obtained results show that both statistical and machine learning approaches are able to capture important consumption patterns and generate reliable forecasts for future electric load values. Model performances were evaluated using standard regression metrics such as Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE).

Overall, this project demonstrates how AI-based forecasting techniques can support more efficient and intelligent energy management systems in real-world scenarios.

2. Methods

2.1 Methodological Overview

This project implements a time series forecasting pipeline for household electricity consumption prediction using the Individual Household Electric Power Consumption dataset from the UCI Machine Learning Repository. The workflow includes data acquisition, preprocessing, exploratory data analysis (EDA), forecasting model development, and evaluation.

The forecasting target is Global_active_power, which represents the total active household power consumption. This variable was selected because it directly reflects electricity

demand and exhibits clear temporal and seasonal behaviour suitable for forecasting applications.

Figure 2.1 summarises the overall forecasting pipeline adopted in the project, from raw data loading to final model evaluation.

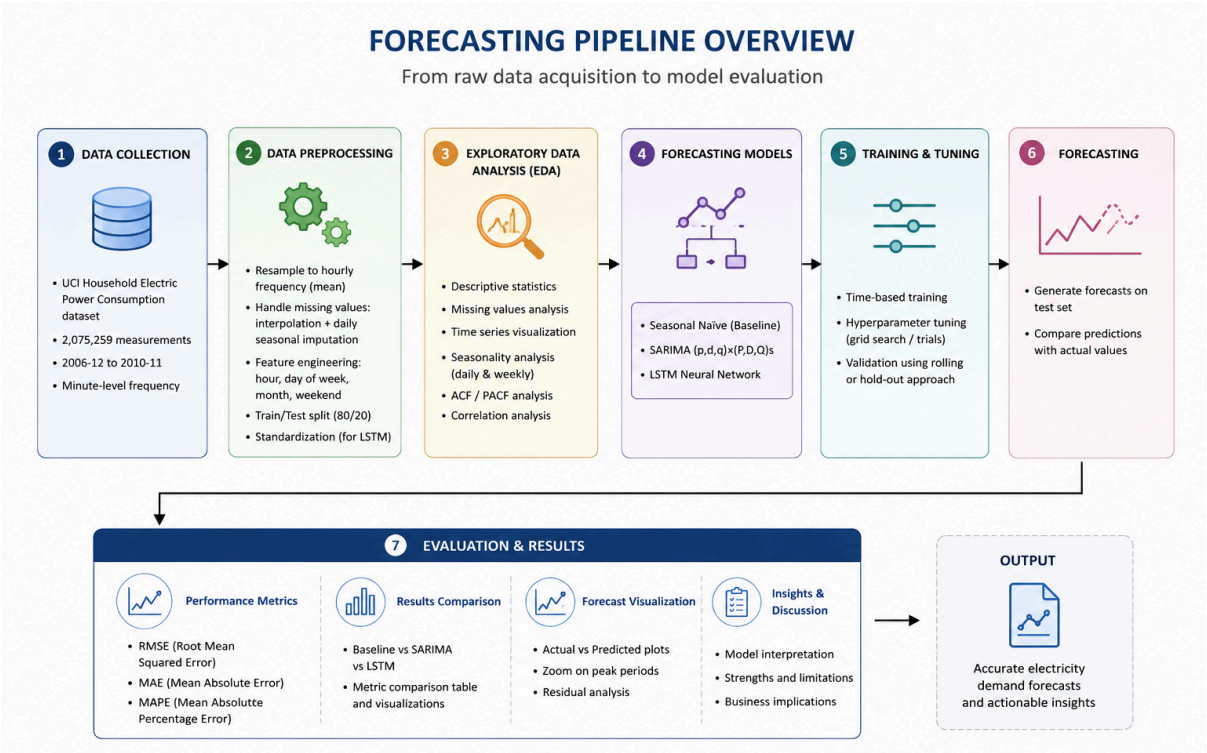

Figure 2.1 *— Forecasting pipeline overview (workflow diagram)*

2.2 Data Collection

The dataset contains minute-level measurements of household electrical consumption collected over several years. The available variables include active power, reactive power, voltage, current intensity, and three sub-metering measurements.

During data loading, the Date and Time columns are merged into a single timestamp index. Missing values represented by the “?” symbol are automatically converted into NaN values, and all numerical variables are cast to floating-point format. The dataset is then sorted chronologically to preserve temporal ordering for forecasting.

The dataset was selected because it provides a long and continuous time series with strong daily and weekly consumption patterns, making it suitable for comparing statistical and neural forecasting methods.

2.3 Data Preprocessing

The original series is recorded at one-minute frequency. To reduce noise and computational complexity, the data are resampled to hourly frequency using mean aggregation. Hourly resolution preserves the dominant daily consumption behaviour while producing a more manageable forecasting dataset.

Missing values are handled using a two-stage imputation strategy:

• Short gaps (up to six hours) are filled using linear interpolation;
• Longer gaps are filled using the value recorded at the same hour on the previous or following day.

This approach was adopted because household electricity demand exhibits strong daily periodicity, making seasonal replacement more reliable for long missing intervals.

After imputation, additional temporal features are extracted from the timestamp index:
• Hour of day;
• Day of week;
• Month;
• Weekend indicator.

The dataset is then divided chronologically into training and testing subsets using an 80/20 split. Random shuffling is intentionally avoided to prevent information leakage from future observations.

For the LSTM model, the target series is standardized using the mean and standard deviation computed only from the training data. This improves optimisation stability while maintaining a fair evaluation procedure.

2.4 Exploratory Data Analysis

Exploratory data analysis was performed to investigate seasonality, temporal dependencies, and correlations within the dataset.

The analysis included:
• Missing-value analysis;
• Seasonal decomposition;
• ACF and PACF analysis;
• Correlation analysis among numerical variables.

The results revealed strong daily seasonality and persistent temporal autocorrelation. These findings motivated the use of hourly resampling, a 24-hour seasonal baseline, and seasonal autoregressive forecasting models.

2.5 Forecasting Models

Three forecasting approaches were implemented and compared.

2.5.1 Seasonal Naïve Baseline

The baseline model predicts future values using the observation from the same hour of the previous day. The seasonal period is therefore fixed at 24 hours.

2.5.2 SARIMA Model

The statistical forecasting model is based on Seasonal ARIMA using the SARIMAX implementation from statsmodels.

The selected configuration is:
• Non-seasonal order: (2,0,1)
• Seasonal order: (1,1,1,24)

SARIMA was selected because it explicitly models temporal autocorrelation and daily seasonal behaviour. Forecasting is performed using a rolling prediction strategy, where the model is sequentially updated with newly observed values. For computational reasons, SARIMA is fitted on the last year (8,760 hours) of the training set.

2.5.3 LSTM Model

The deep learning approach uses a single-layer Long Short-Term Memory (LSTM) neural network implemented in PyTorch.

The model uses a sliding-window forecasting strategy with a lookback window of 48 hourly observations. The architecture consists of one LSTM layer followed by a fully connected output layer.

The main hyperparameters are:
• Hidden size: 64;
• Epochs: 20;
• Batch size: 128;
• Learning rate: 0.001.

The LSTM model was included to evaluate whether nonlinear neural forecasting can better capture complex temporal dependencies compared with traditional statistical methods.

2.6 Evaluation Methodology

The models are evaluated on the held-out test set using aligned timestamps between observed and predicted values.

The evaluation metrics are:
• Root Mean Squared Error (RMSE);
• Mean Absolute Error (MAE);
• Mean Absolute Percentage Error (MAPE).

RMSE is used as the primary metric because it penalises large forecasting errors and preserves the original unit of the target variable. In addition to numerical evaluation, visual diagnostics are used to compare predicted and observed values and to analyse residual behaviour.

3. Results and Discussion

3.1 Exploratory Data Analysis Outcomes

The EDA on the raw dataset [1] confirmed strong daily and weekly regularities.
The boxplots in Figure 3.1 highlight the dominant daily cycle. Electricity consumption reaches minimum values between 02:00 and 05:00 (median below 0.4 kW), shows a moderate morning peak around 07:00–09:00, and a stronger evening peak between 18:00 and 21:00 (median above 1.6 kW). Weekends exhibit a higher daytime profile, while consumption is highest during winter months (December–February), consistent with electric heating and seasonal household behaviour.

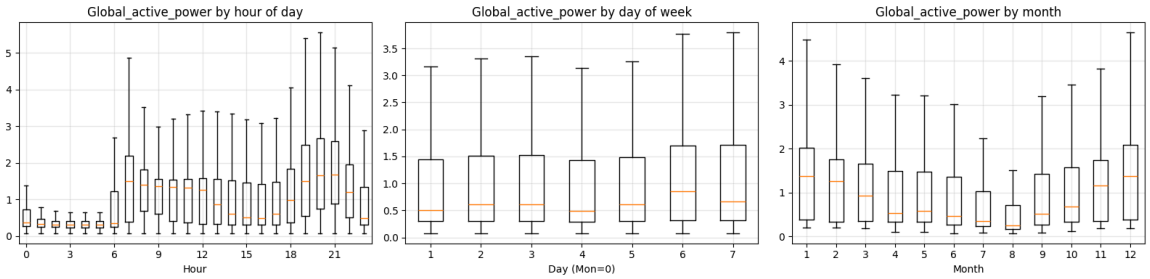

Figure 3.1 *— Distribution of Global_active_power by hour of day, day of week and month.*

Seasonal decomposition with a 24-hour period and the ACF/PACF plots (Figure 3.2) confirm the regularity of the daily cycle: the ACF decays slowly with pronounced peaks at lags 24 and 48, while the PACF cuts off after the first two lags with an additional spike at lag 24. This directly justifies the non-seasonal order (2,0,1) and the seasonal order (1,1,1,24) of SARIMA.

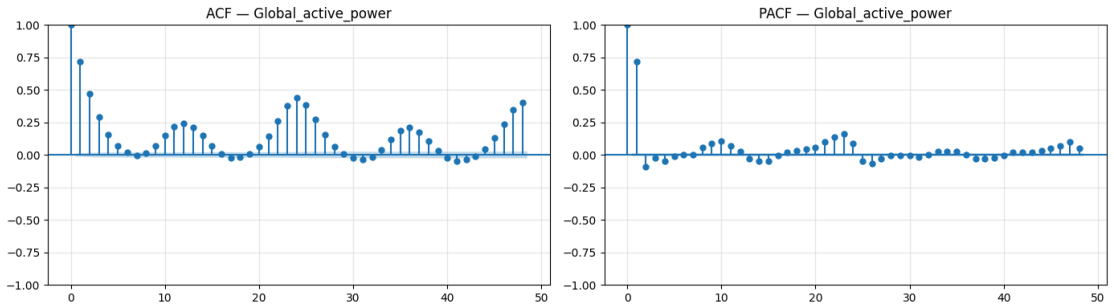

Figure 3.2 *— ACF (left) and PACF (right) of hourly Global_active_power, up to lag 48.*

3.2 Quantitative Model Comparison

The three forecasters were evaluated on the same 6,918-hour test set under an identical day-ahead protocol (stride = 24 hours). The metrics are reported in Figure 3.3.

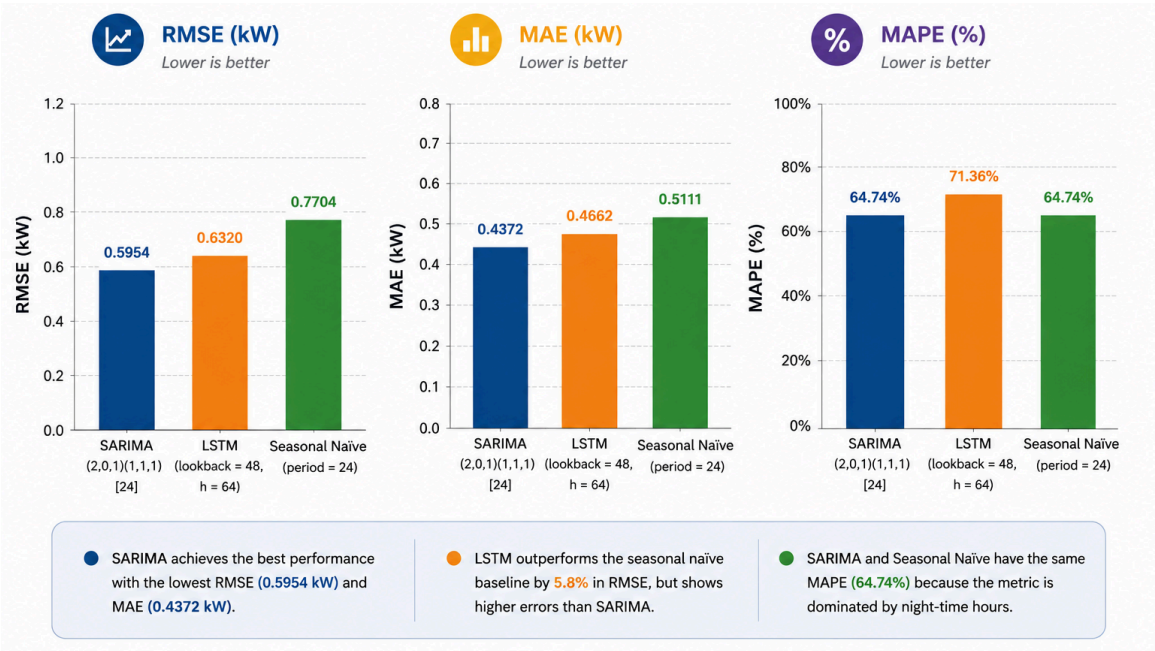

***Figure 3.3** — Comparative performance of SARIMA, LSTM and Seasonal Naïve on RMSE, MAE and MAPE.*

SARIMA achieves the lowest RMSE (0.5954 kW), improving by 22.7% over the Seasonal Naïve baseline [3] (0.7704 kW) and by 5.8% over the LSTM [4] (0.6320 kW). The MAE ranking is consistent, confirming SARIMA as the best-performing model overall.

This result is significant because deep learning approaches are often expected to outperform classical statistical models. However, on this univariate household series, the highly regular seasonal structure is effectively captured by SARIMA.

3.3 Forecast Visualisations

Figure 3.4 overlays the three forecasts on a representative test week (7–13 June 2010). The seasonal-naïve forecast (blue) is rigidly shifted by 24 hours, leading to large errors on atypical days. The LSTM (green) follows the cycle correctly but overshoots evening peaks and lags on morning ramp-up. The SARIMA forecast (orange) is the smoothest and tracks the average consumption level most accurately, although it visibly underestimates the largest evening peaks. The qualitative ranking is fully consistent with Figure 3.3.

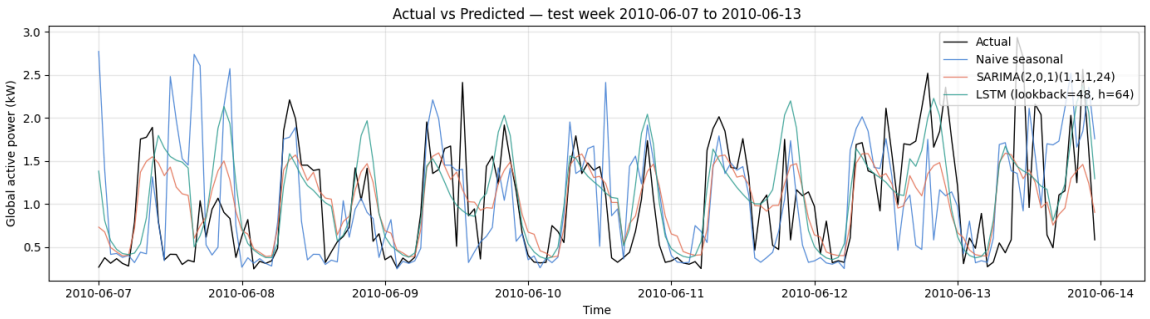

***Figure 3.4** — Actual vs Predicted values on a representative test week (7–13 June 2010).*

3.4 Discussion

Three aspects deserve explicit comment. First, SARIMA outperforms the LSTM because the dominant signal in Global_active_power is the daily seasonal cycle and the short-term autoregressive structure, both effectively captured by SARIMAX [7]. In contrast, the LSTM implemented in PyTorch [8] contains many more parameters and operates on a single noisy series, where additional model complexity provides limited benefits.

Second, the advantage of SARIMA over the Seasonal Naïve baseline (22.7% RMSE reduction) is considerably larger than the improvement over the LSTM (5.8%). This suggests that the main performance gain comes from modelling autocorrelation and seasonality, while nonlinear refinements contribute less significantly. SARIMA therefore represents the best balance between accuracy, interpretability and computational efficiency.

Third, all models tend to underestimate the largest evening peaks and struggle during atypical consumption periods such as holidays or unusual occupancy conditions. Since the models are univariate, they cannot exploit external information that may explain these anomalies. Future improvements could include calendar variables and exogenous features such as temperature.

3.5 Business-Oriented Implications

A SARIMA-based day-ahead forecast with RMSE 0.5954 kW and MAE 0.4372 kW is sufficiently accurate to support short-term planning and resource allocation activities. When aggregated across many households, even small forecasting improvements can reduce dispatching errors and operational inefficiencies at the distribution-network level.

From a grid-stability perspective, all models tend to underestimate evening peaks, potentially increasing balancing costs during high-demand periods. SARIMA reduces this issue compared with the baseline while maintaining lower forecasting variance.

Accurate electricity demand forecasting is also essential for integrating renewable energy sources and supporting sustainable energy management strategies, in line with Enel's focus on decarbonisation and smart-grid optimisation.

Overall, the pipeline delivers reproducible, modular and auditable forecasts that can be deployed in production with limited engineering effort. These forecasts can support data-driven decision-making in smart energy management systems. Future development directions include exogenous regressors, probabilistic forecasts (prediction intervals rather than point estimates), and the extension to multivariate or hierarchical settings where information across many households can be shared.

4. Conclusion

This project developed an AI-based forecasting pipeline for predicting household electricity consumption using historical time series data. Different forecasting approaches were analysed and compared, including Seasonal Naïve, SARIMA and LSTM models.

The results showed that SARIMA achieved the best overall performance, demonstrating that statistical models can effectively capture strong daily and seasonal consumption patterns. The study also highlighted the importance of accurate energy forecasting for improving resource allocation, grid stability and sustainable energy management.

Overall, the project confirms that AI and time series forecasting techniques can support smarter and more efficient energy systems, while also providing a scalable and reproducible framework for future developments.

Bibliography

[1] Hebrail, G., and Berard, A. (2012). Individual Household Electric Power Consumption Data Set. UCI Machine Learning Repository, University of California, Irvine. https://archive.ics.uci.edu/dataset/235
[2] Box, G. E. P., Jenkins, G. M., Reinsel, G. C., and Ljung, G. M. (2015). Time Series Analysis: Forecasting and Control, 5th ed. Wiley, Hoboken, NJ.
[3] Hyndman, R. J., and Athanasopoulos, G. (2021). Forecasting: Principles and Practice, 3rd ed. OTexts, Melbourne. https://otexts.com/fpp3/
[4] Hochreiter, S., and Schmidhuber, J. (1997). Long Short-Term Memory. Neural Computation, 9(8), 1735–1780.
[5] Dickey, D. A., and Fuller, W. A. (1979). Distribution of the Estimators for Autoregressive Time Series with a Unit Root. Journal of the American Statistical Association, 74(366), 427–431.
[6] Kwiatkowski, D., Phillips, P. C. B., Schmidt, P., and Shin, Y. (1992). Testing the Null Hypothesis of Stationarity against the Alternative of a Unit Root. Journal of Econometrics, 54(1–3), 159–178.
[7] Seabold, S., and Perktold, J. (2010). statsmodels: Econometric and Statistical Modeling with Python. In Proceedings of the 9th Python in Science Conference (SciPy), 92–96.
[8] Paszke, A., Gross, S., Massa, F., et al. (2019). PyTorch: An Imperative Style, High-Performance Deep Learning Library. In Advances in Neural Information Processing Systems 32 (NeurIPS).

Appendix A — Code Description

The source code is organised into six self-contained Python modules under the src/ folder, each responsible for one stage of the forecasting pipeline. The execution order is documented in src/README.md.
All modules share global seeds for random, numpy and torch (set via utils.set_global_seed(42)) to guarantee reproducibility.

Module overview

• **data_loader.py** — reads household_power_consumption.txt, parses Date+Time into a single DatetimeIndex, converts '?' tokens into NaN, casts the seven measurement columns to float64, sorts chronologically and returns the resulting DataFrame.
• **preprocessing.py** — resamples to hourly frequency (mean aggregation), handles missing values with a two-stage strategy (linear interpolation for gaps ≤ 6 h, same-hour-previous-day fill for longer gaps), adds calendar features (hour, dayofweek, month, is_weekend) and performs a chronological 80/20 train/test split.
• **eda.py** — exploratory utilities: descriptive statistics, missing-value analysis, temporal-continuity checks, seasonal decomposition (period = 24), ADF and KPSS stationarity tests, ACF/PACF plots, seasonal boxplots and Pearson correlation heatmap.
• **modelling.py** — defines three forecasters under a common interface (fit, forecast_rolling, forecast_rolling_strided): NaiveSeasonalForecaster (period = 24), SARIMAForecaster (SARIMAX from statsmodels, order = (2,0,1), seasonal_order = (1,1,1,24)) and LSTMForecaster (PyTorch, lookback = 48, 1 layer, 64 hidden units).
• **evaluation.py** — aligns observed and predicted values by timestamp, computes RMSE, MAE and MAPE, builds the comparative ranking table and produces the Actual vs Predicted and residual diagnostic plots.
• **main.py** — entry point that orchestrates the full pipeline.

Appendix B — Contribution and AI Statement

Author Contributions

All authors contributed equally to the conceptualisation, software development, analysis, visualisation and writing of this work.
All authors have read and agreed to the submitted version of the report.

Generative AI Statement

In this work, Generative AI tools were used as auxiliary assistants and not as a substitute for the team's design and decision-making. Specifically, we used Anthropic Claude and OpenAI ChatGPT for the following limited purposes: (i) clarification of statistical concepts encountered during the exploratory analysis; (ii) code-level support (i.e. debugging error messages) (iii) language polishing of the technical report, including grammar correction and rephrasing for conciseness.
All AI-generated text and code suggestions were critically reviewed, tested and modified by the authors, who take full responsibility for the content of this report.